

Information Field Theory

A Unified Framework from Relational Information

Single Volume: Foundations, Mathematics, and Extensions

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This book develops the **Information Field Theory** (IFT), a unifying proposal that starts from a strictly positive scalar field $\rho(x) > 0$, interpreted as a density of relational information. From this single primitive field, the known laws of physics — quantum mechanics, general relativity, gauge theory — are derived, together with aspects of quantum biology, consciousness, cosmology, and other disciplines.

The work is structured into three main blocks: the **axiomatic and mathematical foundations** (IFT-Core), which establish the rigorous formalism; the **physical extensions**, including gravity, cosmology, matter, and Yang-Mills theory; and the **interdisciplinary applications**, covering thermodynamics, economics, neuroscience, philosophy, evolution, and climate. Each chapter presents theorems, falsifiable predictions, and, when possible, comparison with recent experimental data (2024-2026).

The central thesis is that information, in its relational sense, is the primary substance of reality, and that all other physical magnitudes — mass, energy, space-time, force — emerge from the geometry and dynamics of the field $\rho(x)$. IFT does not claim to be a closed “theory of everything,” but rather a coherent mathematical framework, open to verification and refutation, capable of guiding research in the coming decades.

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Part I

The Mathematical Core of IFT

Chapter 1

Axiomatic Principles and Fundamental Objects

1.1 The Informational Field

The primitive concept of Information Field Theory is a strictly positive scalar field,

$$\rho : M \rightarrow \mathbb{R}_{>0}, \quad \rho \in C^{2,\alpha}(M) \cap L^1(M), \quad \alpha \in (0, 1), \quad (1.1)$$

defined on an oriented differentiable manifold M of dimension $d = 4$ (Lorentzian space-time, Euclidean space, or lattice). Strict positivity, $\rho(x) > 0$ for all $x \in M$, ensures that the logarithmic transformations and the Fisher metric are globally well defined.

Ontology. We interpret $\rho(x)$ as *density of relational information*: at each point of space-time, ρ encodes the contribution of that point to the global informational flow, together with its relation to the rest of the field. All physics emerges from this single entity.

Derived variables.

- **Informational potential:** $\varphi(x) = \ln \rho(x)$.
- **Informational amplitude:** $u(x) = \sqrt{\rho(x)} = e^{\varphi/2}$.
- **Informational gradient:** $\Xi_\mu(x) = \nabla_\mu \varphi(x) = \frac{\nabla_\mu \rho}{\rho}$.
- **Gradient norm:** $\Xi(x) = \sqrt{g^{\mu\nu} \Xi_\mu \Xi_\nu}$.

All the richness of IFT emerges from the interrelations among these quantities.

Natural scales. From the gradient we define the local length, mass, and frequency scales:

$$\ell(x) = \frac{1}{\Xi(x)}, \quad m_\Xi(x) = \frac{\hbar}{c} \Xi(x), \quad \omega_\Xi(x) = c \Xi(x).$$

These scales connect directly the geometry of information with physics.

1.2 The Eight Axioms of the Core

Axiom 1.1 (Positivity). *There exists a fundamental positive field $\rho(x) > 0$.*

Axiom 1.2 (Relationality). *Physical quantities do not depend on ρ in absolute value, but on its relations, gradients, and curvatures: $\rho, \nabla \rho, \nabla \ln \rho, \nabla \nabla \ln \rho$.*

Axiom 1.3 (General covariance). *The formulation must be invariant under diffeomorphisms of M .*

Axiom 1.4 (Euclidean positivity). *In the Euclidean formulation, the action must be bounded below so that a functional measure $d\mu \sim e^{-S_E} \mathcal{D}\rho$ exists.*

Axiom 1.5 (Controlled locality). *The fundamental action is local or quasilocal, with regularisation terms that guarantee good ultraviolet behaviour.*

Axiom 1.6 (Physical limit). *IFT-Core must recover, as limits, general relativity, scalar field theory, and, through extensions, the Standard Model.*

Axiom 1.7 (Separation of core and extensions). *IFT-Core does not depend on consciousness, biology, or on a particular realisation of the interactions. Those are subsequent modules.*

Axiom 1.8 (Variational principle). *The dynamics is derived from a least-action principle, with a choice of potential that guarantees stability.*

1.3 Configuration Space

The space of physically admissible configurations is

$$\mathcal{F} = \left\{ \rho \in C^{2,\alpha}(M) \mid \rho > 0, \int_M \rho d\mu_g < \infty, \int_M \rho |\nabla \ln \rho|^2 d\mu_g < \infty \right\},$$

endowed with the Fisher-Rao metric

$$\langle \delta\rho_1, \delta\rho_2 \rangle_\rho = \int_M \frac{\delta\rho_1 \cdot \delta\rho_2}{\rho} d\mu_g,$$

which gives it the structure of a Fréchet manifold. The transformation $\rho \mapsto u = \sqrt{\rho}$ establishes a diffeomorphism with the space of positive square-integrable fields.

1.4 Parameters of the Core

The IFT core contains nine free parameters, all dimensional and capable of being determined by experiment or consistency conditions:

Symbol	Meaning	Restriction
G	Gravitational constant	$G > 0$
Λ_0	Bare cosmological constant	free
Z_ρ	Informational kinetic constant	$Z_\rho > 0$
λ	Quartic self-interaction	$\lambda > 0$
v	Vacuum expectation value	$v > 0$
η	Sextic self-interaction	$\eta \geq 0$
κ_ρ	Ultraviolet stabilizer	$\kappa_\rho \geq 0$
λ_F	Fisher deformation	$\lambda_F > 0$
σ	Geometry–information coupling strength	$\sigma > 0$

Compared with the 2 parameters of general relativity or the 19 of the Standard Model, IFT is remarkably economical.

Example. In the decoupled limit ($\sigma \rightarrow \infty$) we recover general relativity with an effective cosmological constant $\Lambda_{\text{eff}} = \Lambda_0 + 8\pi G V(v)$. In the ultralight-field limit ($\Xi \rightarrow 0$), the informational corrections are suppressed and standard physics emerges.

Chapter 2

Fundamental Action and Dynamics

2.1 Minimal Action

In Lorentzian signature $(-, +, +, +)$ the simplest action satisfying the axioms is

$$S_{\text{IFT-Core}} = \int_M d^4x \sqrt{-g} \left[\frac{Z_\rho}{2} g^{\mu\nu} \nabla_\mu u \nabla_\nu u - V(u) \right]. \quad (2.1)$$

In Euclidean formulation (necessary for path integrals) we use

$$S_{\text{Core},E} = \int d^4x_E \sqrt{g_E} \left[\frac{Z_\rho}{2} (\nabla_E u)^2 + V(u) + \frac{\kappa_\rho}{2} (\Delta_E u)^2 \right], \quad (2.2)$$

where the last term, with $\kappa_\rho \geq 0$, acts as an ultraviolet regulator.

2.2 Stable Potential

$$V(u) = \frac{\lambda}{4} (u^2 - v^2)^2 + \frac{\eta}{6} u^6, \quad \lambda > 0, \quad \eta \geq 0. \quad (2.3)$$

For $u = v$ (i.e., $\rho = v^2$) the minimum is reached — the informational vacuum. The addition of the sextic term guarantees global confinement and quantum stability.

2.3 Equation of Motion

Varying the Lorentzian action with respect to u yields

$$Z_\rho \square_g u + V'(u) = 0, \quad (2.4)$$

with $V'(u) = \lambda u(u^2 - v^2) + \eta u^5$. In the regularised Euclidean version,

$$-Z_\rho \Delta u + V'(u) + \kappa_\rho \Delta^2 u = 0. \quad (2.5)$$

These equations govern the fundamental dynamics of the information field.

2.4 Fundamental Fisher Identity

Theorem 2.1 (Fisher Identity). *For any positive field ρ ,*

$$\frac{(\nabla \rho)^2}{\rho} = 4(\nabla u)^2,$$

and therefore the Fisher information coincides, up to a factor of 4, with the kinetic energy of the field u .

Proof. $\rho = u^2 \Rightarrow \nabla \rho = 2u \nabla u$; then

$$\frac{(\nabla \rho)^2}{\rho} = \frac{4u^2 (\nabla u)^2}{u^2} = 4(\nabla u)^2.$$

Integrating over space yields the global identity. □

This theorem is one of the cornerstones of IFT: it establishes that the Fisher information — purely informational — is proportional to the kinetic energy of the field, thereby connecting the notion of information with physical dynamics.

Chapter 3

Informational Geometry

3.1 Local Fisher Tensor

The informational gradient defines a symmetric rank-1 tensor,

$$I_{\mu\nu}(x) = \Xi_\mu(x) \Xi_\nu(x), \quad (3.1)$$

which measures the local direction and intensity of informational variation. It is positive semidefinite and plays the role of a pointwise information metric.

3.2 Hessian of the Dilaton

$$H_{\mu\nu}(x) = \nabla_\mu \nabla_\nu \varphi(x) = \nabla_\mu \nabla_\nu \ln \rho(x). \quad (3.2)$$

This tensor contains the local curvature of the information field: its symmetric part encodes how the informational density accelerates, while its trace is the Laplacian of φ .

3.3 Auxiliary Informational Metric

Combining the Hessian and the Fisher tensor, IFT constructs an auxiliary metric,

$$\boxed{\tilde{g}_{\mu\nu}[\rho] = H_{\mu\nu} - \lambda_F I_{\mu\nu} = \nabla_\mu \nabla_\nu \ln \rho - \lambda_F (\nabla_\mu \ln \rho)(\nabla_\nu \ln \rho)}, \quad (3.3)$$

where $\lambda_F > 0$ is the **Fisher deformation parameter**. This metric is not the physical metric $g_{\mu\nu}$ of space-time, but a geometry derived solely from the informational field. In the limit $\lambda_F \rightarrow 0$ we recover the Hessian; for large λ_F , the direction of maximum gradient is depressed, causing a Lorentzian signature to emerge.

3.4 Lorentzian Signature Theorem

Theorem 3.1 (Local Lorentzian Signature). *Let $p \in M$ be a point where $H_{\mu\nu}(p)$ is positive definite and $\Xi_\mu(p) \neq 0$. There exists a critical value $\lambda^* = h_{\min}/|\Xi|^2$, where $h_{\min}$ is the smallest eigenvalue of $H_{\mu\nu}$, such that for all $\lambda_F > \lambda^*$ the metric $\tilde{g}_{\mu\nu}$ has signature $(-, +, +, +)$ at p .*

Proof. $\tilde{g}_{\mu\nu}$ is a rank-1 perturbation of $H_{\mu\nu}$. By the rank-1 update lemma, the eigenvalue most aligned with Ξ is shifted downward. When λ_F exceeds the ratio of the smallest eigenvalue of H to the squared norm of Ξ , exactly one eigenvalue becomes negative, the others remain positive. The resulting signature is $(-, +, +, +)$. $\square$

Corollary 3.2 (Emergent Time Direction). *The eigenvector associated with the negative eigenvalue aligns with Ξ^μ . Thus, the time direction emerges naturally as the direction of maximum informational gradient:*

$$t^\mu \approx \frac{\Xi^\mu}{|\Xi|}.$$

3.5 Geometry–Information Coupling

The connection between informational geometry and physical geometry is established through an alignment term:

$$S_{\text{align}} = \frac{1}{2\sigma^2} \int_M d^4x \sqrt{-g} \|g_{\mu\nu} - \tilde{g}_{\mu\nu}[\rho]\|^2. \quad (3.4)$$

The parameter σ controls the coupling strength:

- $\sigma \rightarrow \infty$: total decoupling $\Rightarrow$ standard general relativity.
- σ finite: intermediate coupling $\Rightarrow$ informational corrections to gravity.
- $\sigma \rightarrow 0$: strong coupling $\Rightarrow$ physical geometry is completely determined by ρ .

Chapter 4

Canonical Quantisation and Classical Limit

4.1 Canonical Variables

In flat space-time, the momentum conjugate to u is

$$\pi_u(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 u)} = Z_\rho \partial_0 u.$$

Equal-time commutation relations are imposed:

$$[\hat{u}(t, \mathbf{x}), \hat{\pi}_u(t, \mathbf{y})] = i\hbar \delta^{(3)}(\mathbf{x} - \mathbf{y}),$$

and all other commutators vanish.

4.2 Quantum Hamiltonian

$$\hat{H}_\rho = \int d^3x \left[\frac{1}{2Z_\rho} \hat{\pi}_u^2 + \frac{Z_\rho}{2} |\nabla \hat{u}|^2 + V(\hat{u}) + \frac{\kappa_\rho}{2} (\nabla^2 \hat{u})^2 \right]. \quad (4.1)$$

The term with κ_ρ regularises ultraviolet fluctuations and is responsible for the existence of solitons in 3 + 1 dimensions.

Theorem 4.1 (Positivity of the Hamiltonian). *If $V(u) \geq 0$ and $\kappa_\rho \geq 0$, then $\hat{H}_\rho \geq 0$.*

Proof. Each summand is a non-negative operator: $\hat{\pi}_u^2 \geq 0$, $|\nabla \hat{u}|^2 \geq 0$, $V(\hat{u}) \geq 0$ by construction, and $\kappa_\rho (\nabla^2 \hat{u})^2 \geq 0$. $\square$

4.3 Informational Vacuum

The ground state $|0\rangle_\rho$ satisfies $\hat{H}_\rho |0\rangle_\rho = E_0 |0\rangle_\rho$ and is defined via normal ordering : $\hat{H}_\rho : |0\rangle_\rho = 0$. It is the state of minimum energy and maximum informational coherence.

4.4 Informational Quantum Potential

For a scalar field with a phase Φ , $\Psi = ue^{i\Phi/\hbar}$, the amplitude dynamics decouples giving rise to a quantum potential:

$$Q[\rho] = -\hbar^2 \frac{\square u}{u} = -\hbar^2 \left[\frac{1}{4} \Xi_\mu \Xi^\mu + \frac{1}{2} \nabla_\mu \Xi^\mu \right]. \quad (4.2)$$

This potential is purely informational: it depends exclusively on the geometry of ρ .

4.5 Semi-WKB Limit

Theorem 4.2 (Classical Limit). *For coherent states with phase S and slowly varying amplitude,*

$$\Psi_{\hbar} = a_0 e^{iS/\hbar} + \mathcal{O}(\hbar),$$

the phase S satisfies the classical Hamilton-Jacobi equation

$$(\partial_{\mu} S)(\partial^{\mu} S) + m^2 c^2 = 0$$

in the limit $\hbar \rightarrow 0$.

(Sketch of proof: insert the WKB expansion into the Klein-Gordon equation and keep the dominant-order terms.)

Chapter 5

Equivalence with the Lattice Formulation

5.1 Partition Function Theorem

For a hypercubic lattice Λ_a with spacing $a > 0$, the lattice action is

$$S_a[u] = \sum_{x \in \Lambda_a} \left[\frac{Z_\rho}{2} \sum_{\mu} (u_{x+\hat{\mu}} - u_x)^2 + a^4 V(u_x) + \frac{\kappa_\rho}{2a^4} (\Delta_a u_x)^2 \right],$$

where Δ_a is the discrete Laplacian.

Theorem 5.1 (Finite Measure). *If $V(u) \geq -C$ and $V(u) \rightarrow +\infty$ as $u \rightarrow \infty$, the partition function in finite volume is finite:*

$$Z_a = \int_0^\infty \prod_x du_x e^{-S_a[u]} < \infty.$$

Proof. In finite volume there is a finite number of variables u_x . Each one-dimensional integral $\int_0^\infty e^{-a^4 V(u_x)} du_x$ converges because the potential grows confiningly. $\square$

Theorem 5.2 (Continuum Limit). *For $\lambda, \eta > 0$ and $\kappa_\rho \geq 0$, the Euclidean measure $d\mu[u]$ exists as the limit of lattice measures as $a \rightarrow 0$ (after renormalisation).*

Chapter 6

Global Existence and Positivity

6.1 Existence of Classical Solutions

Theorem 6.1 (Global Existence). *For every initial datum $(u_0, \dot{u}_0) \in H^1(\mathbb{R}^3) \times L^2(\mathbb{R}^3)$ with $u_0 > 0$ almost everywhere and finite energy, there exists a unique global solution*

$$u \in C([0, \infty); H^1(\mathbb{R}^3)) \cap C^1([0, \infty); L^2(\mathbb{R}^3))$$

of the equation of motion $Z_\rho \square u + V'(u) = 0$ that preserves energy.

Proof. The system is written as $\partial_t w = Aw + F(w)$ with A generating a unitary group and F locally Lipschitz by Sobolev embeddings. Energy conservation provides a uniform H^1 bound, which prevents blow-up and guarantees global extension by the standard criterion. $\square$

6.2 Strict Positivity

Theorem 6.2 (Preservation of Positivity). *If $\rho_0(x) > 0$ and the initial datum is smooth, then the solution $\rho(x, t) > 0$ for all $t \geq 0$.*

Proof. The equation for $\varphi = \ln \rho$ is $\square \varphi + (\nabla \varphi)^2 + \frac{1}{Z_\rho} V'(e^\varphi) = 0$. If ρ were to attempt to vanish, $\varphi \rightarrow -\infty$ and the term $(\nabla \varphi)^2$ would diverge, forcing a singularity that finite energy prevents. $\square$

6.3 Mass Bound Theorem

Theorem 6.3 (Spectral Mass Bound). *For every excited state $|\psi_n\rangle$ with energy $E_n > E_0$,*

$$m_n = \frac{E_n - E_0}{c^2} \geq \frac{v\sqrt{Z_\rho}}{2\sqrt{2}} \frac{\hbar}{c} \sqrt{\langle \psi_n | \Xi^2 | \psi_n \rangle}. \quad (6.1)$$

Proof. Expand $\hat{u} = v + \hat{\chi}$ around the vacuum and relate the quantum fluctuations of $\hat{\chi}$ with the informational gradient. The Heisenberg inequality $\Delta \hat{u} \Delta \hat{\pi}_u \geq \hbar/2$ provides the lower bound. $\square$

Corollary 6.4 (Positive Gap). *If the vacuum has non-trivial quantum fluctuations ($\langle 0 | \Xi^2 | 0 \rangle > 0$), then the spectrum exhibits a positive mass gap $m_{\text{gap}} > 0$.*

Chapter 7

Topological Classification of Solitons

7.1 Generalised Derrick Theorem

Theorem 7.1 (Derrick with UV Stabilizer). *Let $E[u] = \int [\frac{\kappa_\rho}{2} |\nabla u|^2 + V(u) + \frac{\kappa_\rho}{2} (\Delta u)^2] d^3x$. Under rescaling $u_\lambda(\mathbf{x}) = u(\lambda\mathbf{x})$,*

$$E[u_\lambda] = \lambda^{-1} E_{\text{kin}} + \lambda^{-3} E_{\text{pot}} + \lambda E_{\text{stab}}.$$

The equilibrium condition $\frac{dE}{d\lambda}|_{\lambda=1} = 0$ demands $E_{\text{stab}} = E_{\text{kin}} + 3E_{\text{pot}}$. With $\kappa_\rho > 0$ non-trivial static solutions exist; without it ($E_{\text{stab}} = 0$) we recover the original Derrick theorem, which forbids them.

7.2 Informational Winding Number

Definition 7.2 (Topological Charge). *For configurations that tend to the vacuum, $u(\mathbf{x}) \rightarrow v$ as $|\mathbf{x}| \rightarrow \infty$, the winding number is*

$$n[u] = \frac{1}{4\pi v^2} \int_{\mathbb{R}^3} \epsilon^{ijk} \Xi_i \partial_j \Xi_k d^3x \in \mathbb{Z}.$$

7.3 BPS Bound

Theorem 7.3 (Bogomolny Bound). *For solitons with winding number $n \in \mathbb{Z}$,*

$$M_n \geq \frac{4\pi v}{g} |n|.$$

Equality is attained for solutions satisfying first-order (BPS) equations.

7.4 Complete Classification

Type	Charge	Stabilisation	Physical Example
Topological (π_3)	$n \neq 0$	Conservation of n	Baryon QCD, Skyrmion
Q-ball	$Q > 0$	Noether charge	Scalar dark matter
UV Stabilized	None	$\kappa_\rho > 0$	Glueball 0^{++}
Oscillon	None	Non-linear dynamics	Axion star

Part II

Physical Extensions

Chapter 8

Emergent Gravity

8.1 Gravitational Coupling

The complete gravitational action of IFT is

$$S_{\text{IFT-grav}} = \frac{1}{16\pi G} \int \sqrt{-g} (R - 2\Lambda_0) + S_{\text{IFT-Core}} + S_{\text{align}} + S_{\text{matter}}. \quad (8.1)$$

Einstein's equations are modified by the presence of the informational energy-momentum tensor:

$$G_{\mu\nu} + \Lambda_0 g_{\mu\nu} = 8\pi G (T_{\mu\nu}^{(\rho)} + T_{\mu\nu}^{(\text{align})} + T_{\mu\nu}^{(\text{matter})}).$$

8.2 Newtonian Limit

For static and weak configurations, $u \approx v + \chi$ with $|\chi| \ll v$. The Newtonian potential acquires an informational correction:

$$\nabla^2 \Phi = 4\pi G \rho_{\text{mass}} + \frac{Z_\rho}{2\sigma^2} \nabla^2 \chi^2.$$

The second term represents a fifth force of informational origin.

8.3 Gravitational Waves

Fluctuations of the field ρ along the line of sight produce a phase correction in gravitational waves:

$$h(f) = h_{\text{GR}}(f) \left[1 + \kappa_1 \left(\frac{f}{f_{\text{ref}}} \right) + \mathcal{O}(f^2) \right], \quad (8.2)$$

with $\kappa_1 \sim 10^{-15}$ and $f_{\text{ref}} \sim 100$ Hz. This prediction is testable with third-generation detectors (Einstein Telescope, Cosmic Explorer).

Chapter 9

Dynamic Cosmology

9.1 Friedmann with Informational Field

For a homogeneous and isotropic universe (FLRW), $u = u(t)$. The Friedmann equations are

$$H^2 = \frac{8\pi G}{3}(\rho_m + \rho_r + \rho_\rho) - \frac{k}{a^2} + \frac{\Lambda_0}{3}, \quad (9.1)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho_m + \rho_r + \rho_\rho + 3p_\rho), \quad (9.2)$$

with

$$\rho_\rho = \frac{Z_\rho}{2}\dot{u}^2 + V(u), \quad p_\rho = \frac{Z_\rho}{2}\dot{u}^2 - V(u).$$

9.2 Equation of State

The effective equation of state is

$$w_\rho = \frac{p_\rho}{\rho_\rho} = -1 + \frac{Z_\rho \dot{u}^2}{Z_\rho \dot{u}^2/2 + V(u)}.$$

In the limit $\dot{u} \rightarrow 0$ we recover $w = -1$ (cosmological constant). For slowly rolling fields, w deviates slightly from -1 :

$$w(z) \approx -1 + \varepsilon \frac{z}{1+z}, \quad (9.3)$$

with $\varepsilon \approx 0.12$ – 0.18 for the logarithmic potential $V(u) \propto u^2 \ln(u^2/v^2)$.

9.3 Comparison with Data

DESI DR2 data (2024) show preference for $w \neq -1$ at 2.5σ . The IFT parametrisation is consistent with current observational constraints and will be decisively tested by Euclid (2025-2030) and CMB-S4 (2027-2032).

Chapter 10

Matter and Masses

10.1 Entropic Mechanism for Mass

The mass of an informational soliton is related to its Shannon entropy:

$$M = m_P e^{-S/2}, \quad S[\rho] = - \int d^3x \rho \ln \rho.$$

Compact solitons (low entropy) correspond to massive particles; extended solitons (high entropy) to light particles. This mechanism reproduces qualitatively the mass hierarchy of the Standard Model.

10.2 Informational Higgs

The Higgs field of the Standard Model is reinterpreted as the electroweak component of the informational field:

$$H(x) = u_H(x) \xi_{EW}(x),$$

where $\langle u_H \rangle = v \approx 246 \text{ GeV}$. The masses of gauge bosons and fermions are generated through the informational Yukawa coupling.

Chapter 11

Yang-Mills and the Mass Gap

11.1 IFT-YM Formulation

The Yang-Mills action is regularised with the informational field,

$$S_{\text{YM}}^{\text{IFT}} = \frac{1}{4g^2} \int \text{Tr}(F^2) + \int \left[\frac{Z_\rho}{2} (\partial u)^2 + V(u) + S_{\text{int}}[u, A] \right].$$

The field $u > 0$ acts as a positive regulator, opening a rigorous path towards the solution of the million-dollar problem.

11.2 Progress and Conjectures

- **Lattice existence:** $Z_a < \infty$ in finite volume (proved).
- **Gauge invariance and OS2:** verified.
- **Exponential clustering:** for large m_u , correlators decay exponentially.
- **Conditional chain:** if the uniform continuum limit and the stable decoupling $u \rightarrow v$ are proved, the mass gap follows.

The Yang-Mills mass gap problem remains officially open, but IFT-YM provides a concrete roadmap with rigorous partial advances.

Part III

Interdisciplinary Extensions

Chapter 12

Informational Thermodynamics

12.1 Shannon Entropy and H-Theorem

The informational entropy is

$$S[\rho] = -k_B \int_M \rho \ln \rho d\mu.$$

For gradient-flow dynamics,

$$\partial_t \rho = \nabla \cdot (D \rho \nabla \frac{\delta}{\delta \rho}),$$

the informational H-theorem is proved:

$$\frac{dS}{dt} = k_B \int_M D \rho \left| \nabla \frac{\delta}{\delta \rho} \right|^2 d\mu \geq 0.$$

The second law of thermodynamics emerges as a consequence of informational flow.

12.2 Informational Temperature

$$T_{\text{info}} = \left(\frac{\partial S}{\partial E} \right)^{-1} \sim \frac{\hbar c}{k_B} \frac{1}{L_{\text{info}}}.$$

This relation connects the Hawking temperature of black holes with the informational geometry of the horizon.

Chapter 13

Consciousness as Informational Self-Presence

13.1 Consciousness Functional

Consciousness is operationally defined through the functional

$$\Phi_{\text{IFT}} = \int_{\Omega} dV w(x) \left[\alpha_I \frac{\rho \ln(\rho/\rho_{\text{base}})}{\rho_*} + \alpha_F \ell_F^2 \text{Tr}(\kappa_F) + \alpha_C I_{\text{causal}} \right],$$

together with the dual criterion

$$\Phi_{\text{IFT}} > \Phi_c \quad \wedge \quad \tau \Xi > K.$$

The theory transforms the “hard problem” into a falsifiable empirical programme.

13.2 Experimental Validation

The Cogitate project (2025) showed that the decodability of conscious contents is higher in posterior cortex than frontal, consistent with the IFT prediction of posterior predominance. Studies with anaesthesia, REM sleep, and psychedelics are underway.

Chapter 14

Quantum Biology

14.1 Functional Coherence

IFT explains quantum coherence in biological systems (photosynthesis, enzymes, magnetoreception, olfaction) through the organisation of the bath. The spectral exponent $s = 1 + D_f$ and the relation $\tau_\phi \omega_c \approx K_Q$ have been validated in multiple experiments (2024-2026).

Chapter 15

Other Applications

Applications are presented to economics (crashes as informational phase transitions), neuroscience (experimental protocols for measuring Φ_{IFT}), evolution (fitness as KL divergence), philosophy (informational structural realism), and climatology (tipping points as loss of coherence).

Part IV

Closure and Synthesis

Chapter 16

Universal Master Formula of IFT

$$\rho(x) > 0, \quad u = \sqrt{\rho}, \quad \varphi = \ln \rho, \quad \Xi_\mu = \nabla_\mu \ln \rho$$

$$S_{\text{IFT}} = \int d^4x \sqrt{-g} \left[\frac{R - 2\Lambda_0}{16\pi G} + \frac{Z_\rho}{2} (\nabla u)^2 - V(u) - \frac{\kappa_\rho}{2} (\Box u)^2 + \frac{1}{2\sigma^2} \|g - \tilde{g}[\rho]\|^2 \right]$$

$$m_n \geq C \frac{\hbar}{c} \sqrt{\langle \Xi^2 \rangle_n}, \quad M = m_P e^{-S/2}, \quad \Phi_{\text{IFT}} > \Phi_c \wedge \tau \Xi > K$$

Chapter 17

Final Declaration

Information Field Theory is a unifying framework that takes relational information as the fundamental ontological entity. With only nine parameters, it provides exact theorems, quantitative and falsifiable predictions, and a concrete roadmap towards some of the deepest problems of theoretical physics. It is not presented as a closed theory, but as an open research programme, subject to experimental verification and rigorous critique.

IFT = $\rho(x) > 0$ as the foundation of physical reality

The theory will be strong if it predicts. It will be scientific if it accepts being refuted.
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Appendix A

Master Table of Falsifiable Predictions

Domain	IFT Prediction	Observable	Experiment	Timeframe
Gravitational	$h(f) = h_{\text{GR}}(f)[1 + \kappa_1 f/f_{\text{ref}}]$	$\kappa_1 \sim 10^{-15}$	LIGO/ET/CE	2025-2035
Cosmology	$w(z) = -1 + \varepsilon z/(1+z)$	$\varepsilon \approx 0.15$	DESI/Euclid/CMB-S4	2025-2035
Particles	Dilaton with broad m_u	$g_{u\gamma\gamma} \sim 10^{-12} \text{ GeV}^{-1}$	ALPS II/IAXO	2025-2035
Consciousness	$\Phi_{\text{IFT}} > \Phi_c \wedge \tau\Xi > K$	Φ_{IFT} from EEG/fMRI	Multi-site study	2025-2035
Quantum biology	$\tau_\phi\omega_c \approx K_Q$	τ_ϕ in FMO	2D spectroscopy	2025-2035

Table A.1: Master table of falsifiable predictions